A4 GraphColoring

```
class GraphColoring {
    private int V;  // Number of vertices
    private int[][] graph; // Adjacency matrix
    private int[] colors;  // Colors assigned to vertices
    private String[] colorNames = {"", "Red", "Green", "Blue"}; // Mapping for color codes

    public GraphColoring(int[][] adjacencyMatrix, int numColors) {
        V = adjacencyMatrix.length;
        graph = adjacencyMatrix;
        colors = new int[V]; // Default initialized to 0 (no color)
    }

    // Function to check if it's safe to color a vertex
    private boolean isSafe(int v, int c) {
        for (int i = 0; i < V; i++) {
            if (graph[v][i] == 1 && colors[i] == c) {
                return false;
            }
        }
        return true;
    }

    // Backtracking function to color the graph
    private boolean solveGraphColoring(int v, int m) {
        if (v == V) {
            printSolution();
            return true;
        }
```

```
    for (int c = 1; c <= m; c++) {
        if (isSafe(v, c)) {
            colors[v] = c;
            if (solveGraphColoring(v + 1, m)) {
                return true;
            }
            colors[v] = 0; // Backtrack
        }
    }
    return false;
}

// Function to print the color assignment with names
private void printSolution() {
    System.out.println("Vertex Colors:");
    for (int i = 0; i < V; i++) {
        System.out.println("Vertex " + i + " -> " + colorNames[colors[i]]);
    }
}

// Function to start the coloring process
public void solve(int m) {
    if (!solveGraphColoring(0, m)) {
        System.out.println("No solution exists.");
    }
}

// Main function to run the program
public static void main(String[] args) {
```

```
        int[][] graph = {
            {0, 1, 1, 1},
            {1, 0, 1, 0},
            {1, 1, 0, 1},
            {1, 0, 1, 0}
        };
        int numColors = 3;  // Red, Green, Blue
        GraphColoring gc = new GraphColoring(graph, numColors);
        gc.solve(numColors);
    }
}
```

Output

Vertex Colors:

Vertex 0 -> Red

Vertex 1 -> Green

Vertex 2 -> Blue

Vertex 3 -> Green